22) Show that a set of vectors $\{v_1, v_2, \ldots, v_5\}$ in R^n is linearly dependent when $\dim \text{Span}\{v_1, v_2, \ldots, v_5\} = 4$.

Soln: By defn, $\dim \text{Span}\{v_1, v_2, \ldots, v_5\}$ is the no. of vectors in the basis of $\text{Span}\{v_1, v_2, \ldots, v_5\}$, which is 4.

Let, for the sake of contradiction, $\{v_1, v_2, \ldots, v_5\}$ be linearly independent. By defn, $\{v_1, v_2, \ldots, v_5\}$ is a basis for $\text{Span}\{v_1, v_2, \ldots, v_5\}$.
$\therefore$ By defn, $\text{Span}\{v_1, v_2, \ldots, v_5\} = 5$ which is a contradiction.
$\therefore \{v_1, v_2, \ldots, v_5\}$ is linearly dependent. □

23) If possible, construct a 3×4 matrix A such that $\dim \text{Nul}\, A = 2$ and $\dim \text{Col}\, A = 2$.

Soln: **Claim:** If $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 6 & 10 & 14 \\ -1 & 3 & 2 & 1 \end{bmatrix}$, then $\dim \text{Nul}\, A = 2$ and $\dim \text{Col}\, A = 2$.

Proof:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 6 & 10 & 14 \\ -1 & 3 & 2 & 1 \end{bmatrix} \xrightarrow[-4 \times \text{Row}1]{\text{Row}2 = \text{Row}2} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -2 & -2 & -2 \\ -1 & 3 & 2 & 1 \end{bmatrix} \xrightarrow[+\text{Row}1]{\text{Row}3 = \text{Row}3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -2 & -2 & -2 \\ 0 & 5 & 5 & 5 \end{bmatrix}$$

$$\xrightarrow[\times(-1/2)]{\text{Row}2 = \text{Row}2} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 5 & 5 & 5 \end{bmatrix} \xrightarrow[-5 \times \text{Row}2]{\text{Row}3 = \text{Row}3} \begin{bmatrix} \boxed{1} & 2 & 3 & 4 \\ 0 & \boxed{1} & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The final matrix is in echelon form and its easy to see that Cols 1 and 2 are the only pivot columns. By Theorem 13, the pivot columns 1 and 2 form a basis for Col A. $\therefore$ By defn, $\dim \text{Col}\, A = 2$.
The variables corresponding to Columns 3 and 4 are free variables. By Example 2, $\dim \text{Nul}\, A = 2$. □

24) Construct a 4×3 matrix with rank 1.

Soln: **Claim:** The matrix $\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \\ -1 & -4 & -2 \end{bmatrix}$ has a rank 1.